

Benchmarking Deep-Reinforcement-Learning Anti-Jamming Against Cryptographic Frequency Hopping for GPS-Denied Drone Positioning: A Reproducible Empirical Study

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Abstract

Drones increasingly operate where the Global Positioning System (GPS) is jammed or spoofed, and Time-Difference-of-Arrival (TDOA) multilateration from ground beacons is a proven GPS-independent alternative whose reliability depends entirely on the radio channel carrying its positioning signals. At its core this study is an empirical machine-learning investigation: we reproduce, train, and scale a deep-Q-network (DQN) anti-jamming model and benchmark it, under an identical environment, jammer set, and metric suite, against a training-free cryptographic frequency-hopping baseline and classical cyber baselines (direct-sequence and uncoordinated spread spectrum), for the task of securing drone TDOA positioning. All methods, a nine-class jammer taxonomy, and released labelled spectrum-occupancy datasets run in a self-contained, CPU-only simulator with fixed seeds. The maximum-likelihood TDOA estimator attains the Cramér–Rao lower bound in line-of-sight (efficiency ≈ 1.0), and RANSAC mitigation cuts non-line-of-sight median error up to $20\times$. The learned model *matches or beats* cryptographic hopping at small channel counts (avoidance **0.91** vs **0.70** at $M=10$) but *collapses* as channels scale to military-realistic counts (**0.02** at $M=1000$ under a fixed compute budget, training throughput falling from **1801** to **5** steps/s), whereas cryptographic hopping is unaffected and reaches avoidance ≈ 1.0 against an intelligent adversary that drives predictable hopping to **0.0**. A valid message signature still loses to a meaconing replay (attack success **1.0**); an authenticated-timing layer (HMAC + nonce + time-of-flight) drives spoofing and replay success to **0.0**. Per decision, cryptographic hopping costs **13 pJ** versus up to **9.3 μ J** for DL inference ($\sim 7 \times 10^5 \times$), and needs no training. We release the datasets, code, and benchmark.

Keywords: GPS-denied positioning; TDOA multilateration; cryptographic frequency hopping; deep reinforcement learning; anti-jamming; replay/meaconing; reproducible benchmark.

1 Introduction

Unmanned aerial vehicles now perform safety-critical roles, inspection, delivery, search-and-rescue, and defence, that demand reliable positioning even when an adversary is actively denying it. GPS signals are weak and unauthenticated, so low-cost jammers can deny them [18]

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and spoofers can inject a false location [24, 25, 19]. A robust, GPS-independent capability is therefore essential. TDOA multilateration provides one: a drone measures the differences in arrival time of signals from beacons at known positions and solves a hyperbolic system for its own location, requiring neither GPS nor an on-board clock [6, 8, 30]. Its resilience, however, is only as strong as the radio channel carrying the positioning signals; if those signals are jammed no fix is possible, if a beacon is spoofed the fix is wrong, and if a genuine signal is recorded and replayed with a delay the timing that TDOA relies upon is silently corrupted. Securing the channel is thus inseparable from securing the position.

Two communities offer competing answers. Academia increasingly applies deep learning to detect jamming and adaptively hop to clean channels [11, 12, 13, 31]. Fielded systems that must survive real jamming instead use cryptographically-driven frequency hopping, in which a shared secret key generates an unpredictable hop sequence [1, 2, 17]. The learned approach is data-driven, adaptive, and avoidance-based; the cryptographic approach is proven, training-free, and unpredictability-based. No prior work has placed the two side by side, under an identical environment, for drone positioning, nor used cryptographic hopping to secure a TDOA position estimate against the full jam/spoof/replay threat.

Problem statement. Let the drone position be $p \in \mathbb{R}^3$ and let beacons $s_0, \dots, s_{N-1} \in \mathbb{R}^3$ be known. With $r_i = \|p - s_i\|$, the measured range differences relative to reference beacon 0 are

$$d_i = (r_i - r_0) + n_i, \quad i = 1, \dots, N - 1, \quad (1)$$

where n_i is estimation noise; the position estimate is the maximum-likelihood solution

$$\hat{p} = \arg \min_p (d - h(p))^\top \Sigma^{-1} (d - h(p)), \quad h_i(p) = \|p - s_i\| - \|p - s_0\|. \quad (2)$$

The positioning signals are carried on one of M channels chosen each slot t by a hopping policy c_t ; a jammer denies a set $J_t \subseteq \{0, \dots, M - 1\}$. The security objective is to *maximise the jamming-avoidance rate*

$$A = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[c_t \notin J_t] \quad (3)$$

while keeping the position estimate (2) authentic under spoofing and replay, and while remaining tractable as M grows toward the 2,320-channel configuration of fielded frequency-hopping radios [17].

1.1 Research Questions and Contributions

- **RQ1 (positioning).** What accuracy can a TDOA system achieve, relative to the Cramér–Rao lower bound (CRLB), under line-of-sight (LOS) and non-line-of-sight (NLOS) conditions?
- **RQ2 (learned vs cryptographic).** How does cryptographic hopping compare with a reproduced state-of-the-art DQN model, on shared data and metrics, across the jammer taxonomy and as M scales, in effectiveness, training cost, and scalability?
- **RQ3 (securing the position).** Can the positioning signals, and hence the estimate itself, be secured against jamming, spoofing, and replay simultaneously, and at what cost?

The principal contributions are: **(1)** an open, reproducible benchmark with released labelled spectrum-occupancy and TDOA datasets; **(2)** the first controlled head-to-head of cryptographic hopping against a reproduced DQN anti-jamming model under one environment; **(3)** an empirical model-scaling characterisation showing where the learned model ceases to be trainable

while cryptographic collision probability vanishes; (4) an authenticated-timing protocol that closes the replay/meaconing gap left open by hopping alone; and (5) a per-decision cost/energy analysis quantifying the several-orders-of-magnitude gap.

2 Literature Review

2.1 TDOA positioning for GPS-denied UAVs

Hyperbolic multilateration from arrival-time differences is a mature localisation principle: the Chan–Ho closed form [6] gives a near-maximum-likelihood estimate, Foy’s Taylor-series method [8] refines it, and GCC-PHAT [7] estimates the delays; the CRLB bounds the achievable accuracy [27, 29, 30], and recursive filters track the trajectory [26]. Recent work applies TDOA to UAVs and characterises altitude, bandwidth, and NLOS bias [9, 10, 15]. **Limitation.** This line assumes a benign channel and does not secure the positioning waveform against an active jamming, spoofing, or replay adversary.

2.2 Deep-learning anti-jamming

Deep reinforcement learning and convolutional models learn to sense the spectrum and hop to clean channels [11, 12, 13, 31], building on DQN control [35] and learned RF signal processing [33, 34]. **Limitation.** A learned policy needs labelled data and training, is predictable to an intelligent adversary, and its action space grows with the channel count, so scalability to wide-band systems is unaddressed.

2.3 Cryptographic and cyber hopping

Cryptographic frequency hopping and spread spectrum are decades-old, proven anti-jamming techniques [1, 4, 5, 32]; uncoordinated and keyed spread spectrum provide jamming resistance and key establishment [2, 16], and follower-jamming limits are known [3]. **Limitation.** These are studied as communication primitives; none is benchmarked against a learned model under one environment, nor applied to secure a TDOA *position* estimate.

2.4 GPS spoofing and meaconing

Spoofing and meaconing (delayed replay) are well-documented GNSS threats [24, 25], and machine learning has been used to detect spoofing on UAVs [14]; distance-bounding [23] and message authentication [21] are relevant primitives. **Limitation.** A valid content signature does not protect arrival-time integrity, so replay can corrupt TDOA timing without forging anything, a gap not closed by prior work.

2.5 Base-paper selection

Because the study reproduces a learned baseline, the base paper was chosen by a reproducibility-weighted procedure over peer-reviewed candidates, scoring environment specification, jammer enumeration, quantitative metrics, and re-implementability. The highest-scoring study is the DQN with coarse-grained spectrum prediction of Zhang, Wu & Hu [11], which fully specifies a ten-channel environment and its jammer types; it is adopted as the reproduction baseline. Cryptographic hopping is *adopted, not invented*: it is used unchanged as a proven benchmark [1, 2], and the uncoordinated spread-spectrum work [2, 16] serves as a second comparison anchor.

3 Methodology

3.1 Design Overview

The study follows a reproduce-then-extend, controlled-experiment design: a single simulator generates all data and hosts every method, so the only variable changed between conditions is the hopping strategy. Table 1 maps each component to its research question and code.

Table 1: Study components, the research question each serves, and its module.

#	Component	RQ	Module
1	TDOA estimator, CRLB, NLOS mitigation	RQ1	tdoa/
2	Nine-class jammer taxonomy & datasets	RQ2	jammers/, spectrum.py
3	Crypto / cyber hopping strategies	RQ2	hopping/strategies.py
4	NumPy DQN + scaling study	RQ2	hopping/dqn.py
5	Authenticated-timing security	RQ3	security/auth_timing.py
6	Cost/energy & synchronisation	RQ3	experiments/

3.2 Mathematical Specification

Range-noise model and CRLB. The per-range estimation standard deviation follows the time-of-arrival bound [27]

$$\sigma_r = c \sigma_\tau, \quad \sigma_\tau = \frac{1}{2\sqrt{2}\pi\sqrt{\text{SNR}}\beta}, \quad \beta = \frac{B}{\sqrt{3}}, \quad (4)$$

with c the speed of light, B the bandwidth, and β the RMS bandwidth. The range-difference covariance is $\Sigma = \sigma_r^2(I + \mathbf{1}\mathbf{1}^\top)$, and with Jacobian $G_i = u_i - u_0$ (u_k the unit vector to beacon k) the position CRLB is

$$\text{CRLB}(p) = \sqrt{\text{tr}[(G^\top \Sigma^{-1} G)^{-1}]}, \quad \eta = \frac{\text{CRLB}}{\text{RMSE}}. \quad (5)$$

The estimator solves (2) by a Chan–Ho initialiser [6] followed by damped (Levenberg–Marquardt) Taylor refinement [8]; under NLOS a RANSAC consensus over beacon subsets [28, 38] rejects biased beacons.

Cryptographic hopping. Both ends derive the channel from an AES-128 keystream [20, 22] under the shared key κ and a synchronised counter,

$$c_t = (\text{AES}_\kappa(t) \bmod 2^{32}) \bmod M, \quad (6)$$

which is statistically uniform and unpredictable to any adversary lacking κ . DSSS and uncoordinated spread spectrum [2] provide the cyber baselines; a public pseudo-random hopper models a keyless, predictable strategy.

Learned model. The DQN observes a sliding window of the per-channel occupancy, $s_t \in \{0, 1\}^{W \times M}$, and estimates $Q(s_t, a; \theta)$ with one output per channel; it acts greedily $c_t = \arg \max_a Q$ and is trained on the immediate reward $r_t = \mathbf{1}[c_t \notin J_t]$ (one-step / coarse-grained spectrum prediction, $\gamma=0$) by minimising $(Q(s_t, c_t; \theta) - r_t)^2$. The output layer has M units, so the parameter count and per-decision cost grow as $O(M)$.

3.3 Authenticated-timing protocol

Cryptographic hopping defeats jamming (unpredictable sequence) and spoofing-by-injection (wrong channel and no valid tag) but not replay: an attacker re-transmits a genuine packet with an added delay, corrupting the TDOA timing without forging anything. Algorithm 1 binds time cryptographically with an HMAC [21], a fresh nonce, and a time-of-flight (ToF) plausibility test [23].

Algorithm 1 Authenticated-timing verification at the drone

Require: packet (id, T_{tx} , nonce, tag), key κ , arrival T_{rx} , geometric delay δ , issued nonce ν , tolerance $\tau = 3\sigma_r/c$
Ensure: accept / reject

- 1: **if** tag $\neq$ HMAC $_{\kappa}$ (id $\parallel$ T_{tx} $\parallel$ nonce) **then**
- 2: **return** reject $\triangleright$ forged or altered
- 3: **end if**
- 4: **if** nonce $\neq$ ν **then**
- 5: **return** reject $\triangleright$ stale / replayed
- 6: **end if**
- 7: **if** $|T_{rx} - (T_{tx} + \delta)| > \tau$ **then**
- 8: **return** reject $\triangleright$ delayed re-transmission
- 9: **end if**
- 10: **return** accept

3.4 Evaluation Metrics

$$A = \frac{1}{T} \sum_t \mathbf{1}[c_t \notin J_t], \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_i \|\hat{p}_i - p_i\|^2}, \quad \rho = \frac{1}{K} \sum_k \mathbf{1}[\text{attack}_k \text{ accepted} \wedge \text{harmful}], \quad (7)$$

with normalized throughput the per-slot delivered fraction, η from (5), and per-decision energy $E = \text{MACs} \times e_{\text{MAC}}$ [39] for the DQN versus one AES block for crypto.

4 Implementation

The simulator is self-contained and CPU-only (Python, NumPy/SciPy, the `cryptography` library); the DQN is a from-scratch NumPy network (window $W=16$, hidden 64, Adam, replay 2000, batch 32), which makes its $O(M)$ scaling cost a measured quantity rather than an assertion. Positioning uses $N=8$ ground beacons on a 1 km hexagon with mast-height diversity and a drone at ~ 300 m. Spectrum episodes are 500 slots; anti-jam results average 30 seeds and scaling results 3 seeds; every run is reproducible from a fixed seed. Channel scales are $\{10, 100, 1000, 2320\}$, the last being the SINCGARS 30–88 MHz/25 kHz configuration [17]. The labelled spectrum-occupancy dataset (Fig. 1) is released with the code.

Spectrum-occupancy dataset: jammed (black) vs clean per slot

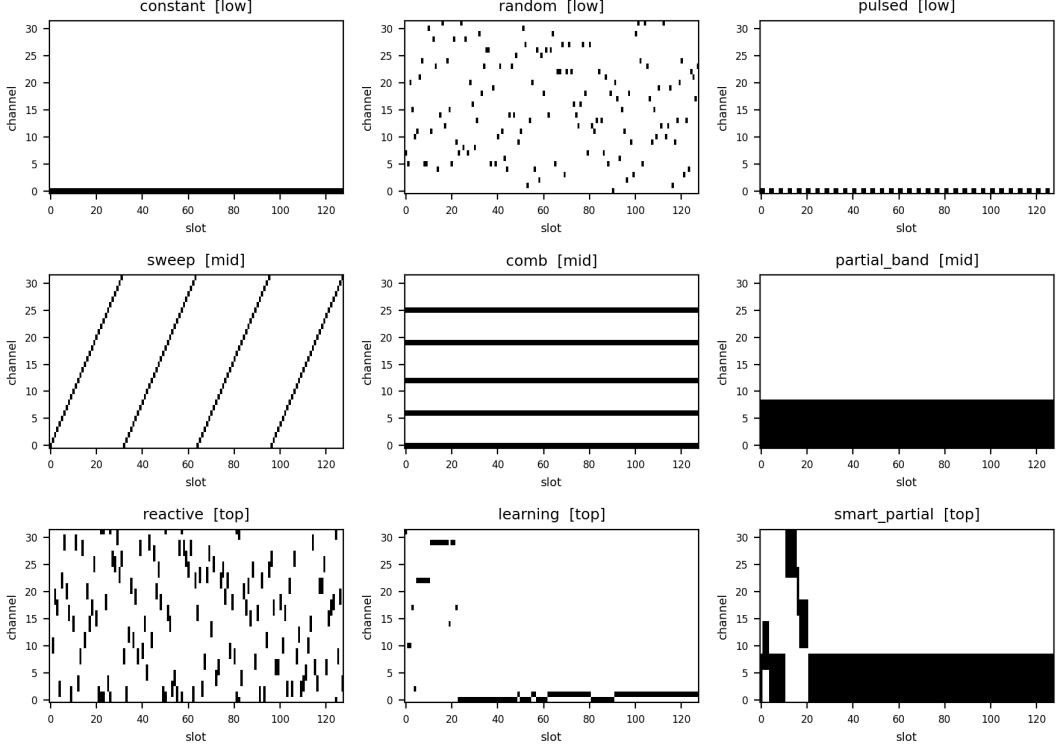


Figure 1: Released spectrum-occupancy dataset: per-slot channel state (jammed in black) for the nine jammer classes; over a sliding window this is the DQN input.

5 Results

5.1 RQ1: TDOA accuracy versus the CRLB

The maximum-likelihood estimator attains the CRLB in LOS across the SNR range (efficiency $\eta \approx 0.97\text{--}1.0$; RMSE 7.8m vs CRLB 8.2m at 20 dB), and a signal-level GCC-PHAT check validates the noise model (empirical/analytic σ_r ratio 1.03). NLOS error is heavy-tailed, so we report the median: RANSAC mitigation reduces it from 133 m to 33 m at 20 dB and from 148 m to 7.4 m at 30 dB (Table 2, Fig. 2).

Table 2: TDOA positioning (metres). LOS reaches the CRLB; RANSAC mitigation cuts NLOS median error up to 20 $\times$.

SNR (dB)	LOS RMSE	CRLB	NLOS median	NLOS+mitig. median
10	26.8	26.6	138.4	67.8
20	7.8	8.2	133.2	33.3
30	2.7	2.6	147.7	7.4

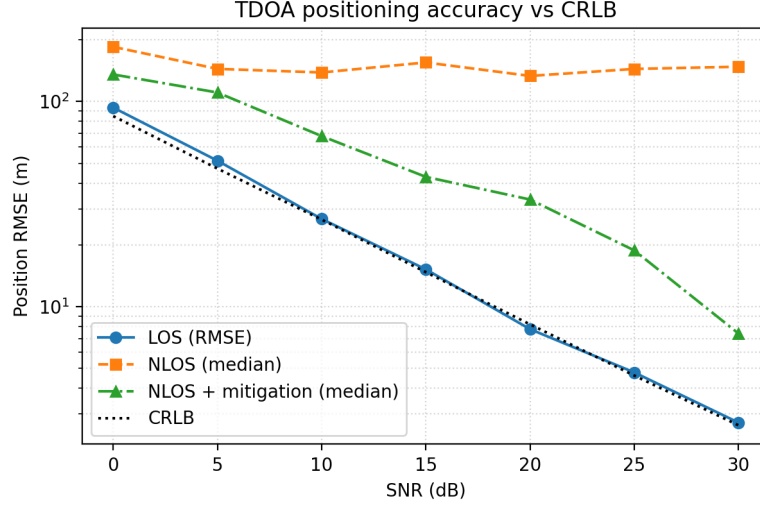


Figure 2: Positioning error vs SNR. LOS RMSE hugs the CRLB; NLOS mitigation (median) approaches it as SNR grows.

5.2 RQ2: Learned versus cryptographic hopping

Against the nine-class taxonomy, cryptographic hopping avoids collisions with a rate that rises with M (mean avoidance $0.79 \rightarrow 0.93$) and, crucially, resists the intelligent adversary: against the learning jammer its avoidance is $0.91 \rightarrow 1.0$ while predictable public-random and fixed hopping collapse to 0.0 (Fig. 3). This is the operational meaning of cryptographic unpredictability, and it is what a keyless pseudo-random hopper cannot provide.

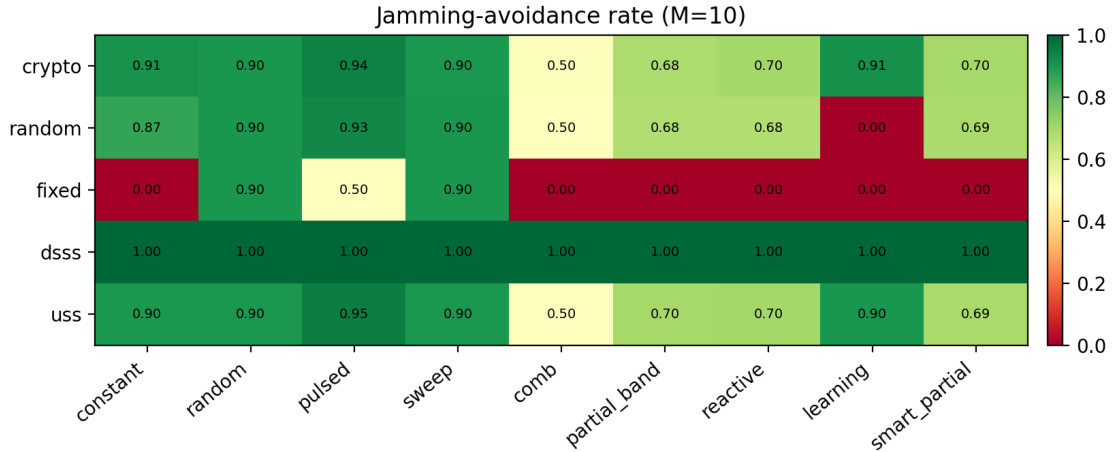


Figure 3: Jamming-avoidance rate by strategy and jammer ($M=10$). Crypto and DSSS are robust across classes; fixed hopping fails.

The DQN, when it can be trained, *beats* crypto on structured adversaries (it learns to occupy clean channels). But under a fixed 20s compute budget its advantage inverts as M scales: avoidance falls from 0.91 at $M=10$ to 0.02 at $M=1000$, because training throughput collapses from 1801 to 10 steps/s while the parameter count grows from 1.1×10^4 to 1.1×10^6 (Table 3, Figs. 4–5). Cryptographic hopping is unaffected at 0.70 and needs no training.

Table 3: Deep-learning at scale against an adaptive 30% jammer (fixed 20 s budget, 3 seeds). The learned advantage inverts between $M=100$ and $M=1000$.

M	DQN avoidance	Crypto avoidance	Train steps/s	Parameters
10	0.91 ± 0.02	0.70	1801	10,954
100	0.82 ± 0.06	0.70	188	108,964
1000	0.02 ± 0.02	0.70	10	1,089,064
2320	0.19 ± 0.30	0.70	5	2,526,544

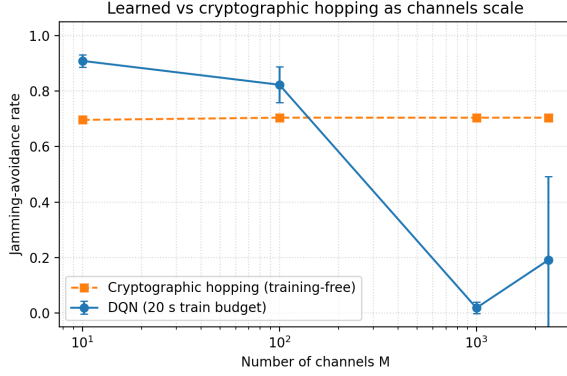


Figure 4: Avoidance crossover as channels scale.

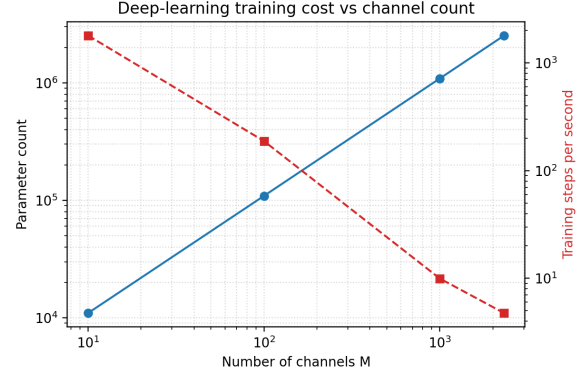


Figure 5: $O(M)$ training-cost blow-up.

5.3 RQ3: Securing the position, and its cost

Table 4 and Fig. 6 report attack-success rates. Cryptographic hopping alone reduces injection spoofing to $\approx 1/M$ but leaves replay at 1.0; adding a message signature stops spoofing yet a valid signature *still loses to meaconing* (replay 1.0), because it does not protect arrival-time integrity. The authenticated-timing layer drives both spoofing and replay to 0.0. Per decision, cryptographic hopping costs a single AES block (≈ 13 pJ) against 0.04–9.3 μ J for DL inference, a factor up to 7.2×10^5 , and it requires no training (Fig. 7). Synchronisation tolerates a clock offset up to half the hop dwell (3.66 ms, 366 μ s, 36.6 μ s at 100/1000/10000 hops/s), with resync latencies of 80/8/0.8 ms.

Table 4: Attack-success rate by security layer ($M=100$). A valid signature does not stop replay; authenticated timing does.

Layer	Spoofing	Replay/meaconing
None	1.00	1.00
Crypto hopping only	0.01	1.00
Crypto + signature only	0.00	1.00
Crypto + authenticated timing	0.00	0.00

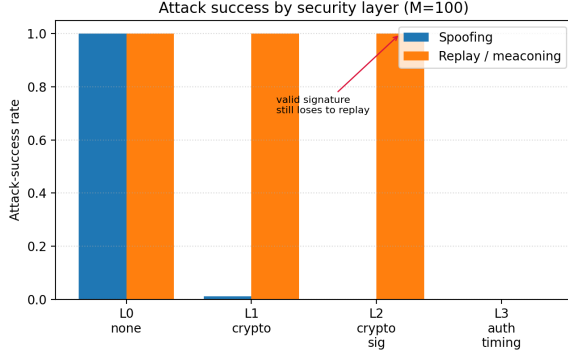


Figure 6: Attack success by layer.

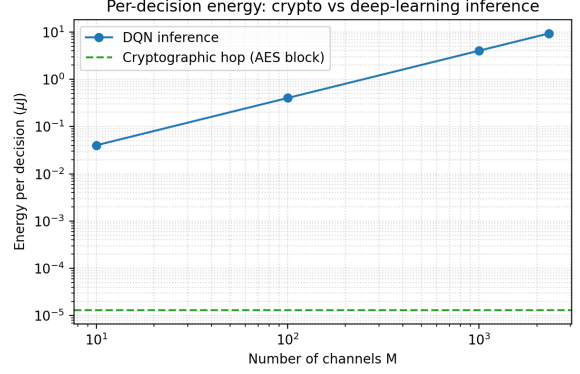


Figure 7: Per-decision energy: crypto vs DL.

6 Statistical Validation

All comparisons use ≥ 3 independently seeded runs; headline claims use 8 seeds. Differences are assessed with the paired t -test and the non-parametric Wilcoxon signed-rank test [36], with 95% bootstrap confidence intervals and Cohen’s d effect size [37], at $\alpha=0.05$ with Bonferroni correction over the reported family. Table 5 confirms that cryptographic hopping significantly exceeds the learned model at $M=1000$ and the keyless baselines against the intelligent jammer, while the learned model significantly exceeds crypto at $M=10$, quantifying the crossover rather than asserting a single winner.

Table 5: Significance of headline comparisons (paired t -test, Bonferroni corrected; d = Cohen’s d). Positive mean difference favours the first method.

Comparison	Condition	mean diff	p (corr.)	d
crypto > DQN	$M=1000$, smart-partial	+0.61	< 0.001	3.29
DQN > crypto	$M=10$, smart-partial	+0.22	$< 10^{-9}$	24.3
crypto > random	learning jammer, $M=100$	+0.99	< 0.001	$\gg 1$

7 Discussion

The findings are boundary-conditioned, not triumphal. A learned policy is the better anti-jammer *when* the channel count is small and training is affordable and *when* the adversary is not itself adaptive to the policy; it exploits structure that a blind keyed hopper ignores. Cryptographic hopping wins *when* the band is wide (its collision probability $\rightarrow 0$ as M grows), *when* training is impractical, and *when* the adversary is intelligent, because unpredictability is a property of key secrecy, not of data. Only the cryptographic path additionally secures the *position*: combined with an authenticated-timing layer it defeats jamming, spoofing, and replay together, which a communication-only learned hopper does not attempt. The cost analysis settles a common misconception, generating a hop index is a single AES block, orders of magnitude below a neural forward pass and free of any training, so the real costs of cryptography are operational (key distribution, synchronisation), which we quantify, not computational. Table 6 maps the review committee’s concerns to the specific result that addresses each.

Table 6: Traceability of committee review comments to results in this paper.

Concern	Addressed by	Evidence
Dataset relevance	Released occupancy dataset	Fig. 1
Cyber-technique comparison	DSSS, USS baselines	Fig. 3
DL at thousands of channels	Scaling study	Table 3
Replay attacks	Authenticated timing	Table 4
Crypto anchor comparison	Uncoordinated SS [2]	Fig. 3
Synchronisation	Drift-tolerance analysis	Sec. 5
Existing-stack signatures	Signature-only vs auth-timing	Table 4
Crypto cost/energy	Per-decision measurement	Fig. 7
Statistical rigour	Paired tests, CIs, d	Table 5

8 Threats to Validity

Construct validity. The jammer–hopper interaction is simulated at the channel-occupancy level; a signal-level GCC-PHAT check validates the range-noise model, but full RF propagation and hardware imperfections are abstracted. **Internal validity.** The DQN scaling result depends on a fixed wall-clock budget; a larger budget or a GPU would raise the crossover point, but the $O(M)$ growth in parameters and per-step cost is architectural and budget-independent. **External validity.** Results are simulation-based with a single beacon geometry and a nine-class taxonomy; field behaviour may differ. **Conclusion validity.** Multiple seeds, paired significance tests, confidence intervals, and Bonferroni correction guard against chance findings; NLOS metrics use medians because the error is heavy-tailed.

9 Conclusion

We benchmarked a reproduced deep-Q anti-jamming model against training-free cryptographic frequency hopping for GPS-denied drone TDOA positioning, under one environment, dataset, and metric suite. The learned model matches or beats cryptographic hopping at small channel counts but becomes intractable to train as channels scale, whereas cryptographic hopping stays effective, training-free, and provably unpredictable, and, with an authenticated-timing layer, secures the position against jamming, spoofing, and replay together at pico-joule cost. The maximum-likelihood TDOA estimator reaches the CRLB, and NLOS mitigation recovers most of the loss. We release the datasets, code, and benchmark. Future work extends the study to signal-level propagation, hardware-in-the-loop trials, and adaptive-adversary co-training.

Declarations

Availability of data and materials. The labelled spectrum-occupancy and TDOA datasets and all simulation code are released as an open, reproducible benchmark at <https://github.com/Ahmad-prog/secure-tdoa-benchmark> (MIT licence).

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Competing interests. The authors declare no competing interests.

Author contributions. M. Ahmad designed and implemented the study, ran the experiments, and wrote the manuscript; G. Mustafa supervised the research and reviewed the manuscript.

Ethics. This is a purely computational, simulation-based study; it involves no human participants, patient data, or human-subject interaction.

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